

DIRECTORY

With the Enlightenment came a focus on reason, skepticism, and liberty that encouraged the systematic exploration of theories, and science became a distinct academic discipline. Increasingly sophisticated equipment, methods, and proofs emerged, as did some of the first modern scientific institutions.

MARCELLO MALPIGHI

1628–1694

In about 1661, while studying frog lungs under a microscope, Italian physician and biologist Marcello Malpighi spotted capillaries—tiny blood vessels that link arteries and veins. In the course of his work, he developed new methods for studying microscopic things, such as illuminating specimens or staining them so they could be seen more easily. His pioneering work in microanatomy led to him becoming an honorary member of the Royal Society in 1669, but Malpighi continued to practice medicine throughout his career. He became the pope's physician in 1691.

JAN SWAMMERDAM

1637–1680

Born in Amsterdam, the Netherlands, Jan Swammerdam qualified as a doctor but dedicated his life to the microscopic study of insects. His main achievements were his discovery of red blood cells and also metamorphosis, which he proved by dissecting an egg, larva, pupa, and adult insect under the microscope and showing that they were simply different forms of the same animal. Swammerdam went on to divide insect development

into four main types, three of which exist in modern classification. But his successes did not impress his father, who withdrew his support. Destitute and depressed, Swammerdam died of malaria aged 43.

NICOLAS STENO

1638–1686

Despite a brief scientific career, Nicolas Steno made major discoveries in anatomy and geology. The Danish scientist likened the teeth of sharks to the “tongue stones” found in Cenozoic rocks. This led him to wonder how fossil shark teeth could have become embedded in rock. Steno went on to establish the underlying principles of how strata (rock layers) form, explaining that the lower the level, the older the rock, and that strata begin as horizontal layers that are disrupted over time. In 1667, he abandoned science for religion and became a bishop 10 years later.

JACOB BERNOULLI

1655–1705

Defying his father's wishes by studying mathematics, Jacob Bernoulli became the first of several famed mathematicians in the Swiss Bernoulli family. He became

professor of mathematics at the University of Basel, Switzerland, in 1687, and taught his younger brother Johann. The brothers worked together on various ways of using calculus, but soon became rivals. Bernoulli is best known for his work in probability and especially for his law of large numbers.

DANIEL BERNOULLI

1700–1782

Jacob Bernoulli's brother Johann had a son, Daniel Bernoulli, who studied philosophy, logic, and medicine and was taught mathematics by his father. In 1738, he published *Hydrodynamica*, which included his discovery that if a stream of moving fluid (liquid or gas) speeds up, its pressure drops. This—Bernoulli's principle—is one of the cornerstones of modern aerodynamics. Bernoulli won 10 awards from the Paris Academy of Sciences in his lifetime, one of which he shared with his father, who—frustrated not to be the sole winner—threw his son out of the house.

EMILIE DU CHATELET

1706–1749

A talented physicist and mathematician, Emilie du Châtelet is also remembered by history for being Voltaire's mistress. Born into a wealthy family in Paris, France, du Châtelet learned science and mathematics from a young age, as well as six languages. Her father, a court official, included her when famous scientists visited their home. She wrote